

Discriminating Earthquakes and Explosion Events by Seismic Signals Basing on BP-Adaboost Classifier

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Abstract—Aiming to discriminate earthquakes and explosion events by seismic signals, this paper makes comparisons among Back-propagation neural networks(BP-NN), support vector machine(SVM), and an ensemble learning method: BP-Adaboost. Experiments on our wave features dataset extracted from seismic signals of earthquake and explosion events, have shown that BP-Adaboost can achieve the overall correct recognition rate not less than 98%, with excellent generalization capability, obviously surpassing BP-NN and SVM. The explanation of BP-Adaboost superiors to other two classical main methods is also briefly given.

Keywords—classification; seismic wave signal; BP-adaboost; ensemble learning; BP neural networks; support vector machine(SVM)

I. INTRODUCTION

Being originated from natural earthquakes and artificial explosion events, the seismic wave signals can be recorded by nearby or ever remote seismic network. How to quickly and accurately discriminate earthquake and explosion event by seismic signals, namely to rapidly identify seismic source type, is very significant importance for national interests and people benefits, and also for world peace. It has always been a central problem in seismology to identify source type by seismic signals. In the beginning, seismologists used statistical methods to infer the classification of earthquakes and explosions [1], the inference was possible under the conditions: the occurrences of explosions were within particular time ranges and within some specific geographic regions, while the occurrences of earthquakes were rather random despite of time ranges or geographic regions. Obviously, this primitive method is heavily affected by underlying dataset with low repeatability. In order to solve this problem, some researchers have already utilized neural network(NN): Genetic BP-NN for identifying the earthquake and blast [2], multilayer perceptron neural network [3] for classifying underwater explosion and earthquake(92% overall correct recognition rate), Benbrahim [4] has proved that the neural network was robustness and flexibility for classifying seismic signals. However neural networks have intrinsic defects in classification, such as: prone to be locked in local optimum, slow convergent speed. Support vector machine (SVM) has been become the main forces for earthquake and explosion classification due to rapid calculation speed and high correct recognition rate: Hanming

Huang [5] used wavelet features and SVM classifier, Mingxia Bi [6] used HHT transformed features and SVM classifier, in those investigations, more than 90% correct recognition rates had been reported. Kernel functions cannot be effectively chosen is the main obstruct for the application of SVM. Not suitable to multi-classes classification problem is another limitation of SVM.

More fresh earthquake and explosion classification methods ang [7], [8] have continually been devised, but no comprehensive performance comparisons among these methods have yet been conducted. In this paper, we proposed a BP-Adaboost method for earthquake and explosion classification problem, which being combined with ensemble learning methods. We compared the performance of BP-Adaboost, BP-NN and SVM in the same dataset. Through the comparisons, we found that BP-Adaboost has higher accuracy and faster speed of recognition, and has better generalization capability, suitable to earthquake and explosion classification. This paper provides an efficient and reasonable new options for the classification of earthquakes and explosion events by seismic wave signals.

This paper is organized as follows. Section 2 presents an introduction of ensemble learning methods and the BP-Adaboost method. The following section gives experimental results and makes respective analysis. The last section summarizes the works.

II. ENSEMBLE LEARNING METHODS

A. Introduction of Ensemble Learning Methods

"Two heads are better than one", the essence of ensemble learning and the old saying is coincide: by firstly producing a lot of simple, better only than random guess weak classifiers, then combining the multitudinous "weak classifiers" according to a certain rules to form a single "strong classifier", would produce perfect classification results. As a promising method in machine learning field, ensemble learning has achieved tremendous success in many applications: text classification, disease diagnosis, computer vision, [13]-[15] etc. If being compared with the BP-NN and SVM method, ensemble learning has many advantages: reducing variance and bias simultaneously, better generalization capability[9], preventing overfitting problem, arbitrarily combining of "weak classifiers" to meet the needs of the various practical problems, and thus better practical problems solving capabilities.

There are three kinds of main ensemble learning methods: Bagging algorithm, Boosting algorithm and Stacking algorithm. The structural diagram of ensemble learning method is shown in Fig. 1.

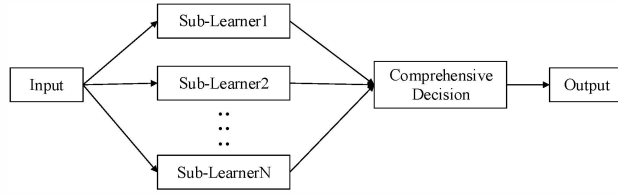


Figure 1. Structure diagram of ensemble learning method

B. BP-Adaboost Method

BP-Adaboost method takes BP-NN as a "weak classifier", then uses Adaboost method combining into a "strong classifier". Due to Adaboost method using greedy strategy, hence does not need to consider the form of weak classifier, and the algorithm itself can adapt to the "weak classifier" respective training error[9], therefore, this paper selects BP-NN as weak classifier. The BP-Adaboost method is shown in Fig. 2.

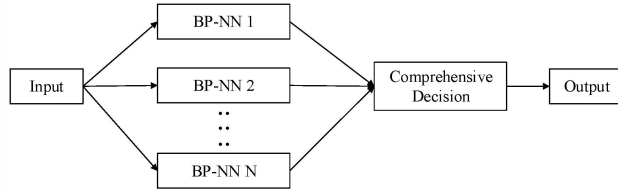


Figure 2. Structure diagram of BP-Adaboost method

Algorithm 1 -- The BP-Adaboost Procedure:

Algorithm 1: BP-Adaboost Algorithm

Input: Training Set $\mathbf{X} \in \mathbf{R}^{d \times n}$

Output: Classification Accuracy

1.Data Selection: randomly select M set of training data, initialize weights distribution as follows:

$$\Theta_{i(1)} = 1 / m \quad (1)$$

2.Initializing BP network:make the network settings as simple as possible, to simplify model;

3."Weak Classifiers"Training: when train the t^{th} "weak classifier",get the predict sequence $g(t)$ of sum of the predict error from BP network:

$$e(t) = \sum_i \Theta_{i(t)}, i = 1, 2, \dots, m(g(t) \neq y) \quad (2)$$

4.Calculating the predicted sequence weights a_t :

$$a_t = \frac{1}{2} \ln \left(\frac{1 - e^t}{e^t} \right) \quad (3)$$

5. Weight adjustment: adjust the weight of the next round according to the iteration:

$$\Theta_{i(t+1)} = \frac{\Theta_{i(t)}}{B_i} * \exp(-a_t y_i g_t(x_i)), i = 1, 2, \dots, m \quad (4)$$

6. "Strong Classifier"output: combined with T "weak

classifier"function $f(g_t, a_t)$ can get the "strong classifier"output:

$$h(x) = \text{sign} \left[\sum_{t=1}^T a_t * f(g_t, a_t) \right] \quad (5)$$

III. EXPERIMENTAL RESULTS AND ANALYSIS

A. Experimental Data

Experimental dataset are from China earthquake science data center. Totally 656 seismic wave signals[5][6], in that 138 earthquake wave signals, 518 explosion wave signals, are selected. From each seismic wave signal, 27 features are extracted, these features are time domain features (standard deviation, mean, median, maximum, minimum, kurtosis and skewness), wavelet features(energy ratio, shannon entropy, logarithmic energy ratio), the Ap1, Asm, Apm, Ap1/Asm, Apm/Asm, Apm/Tme, Asm/Tme, Apm/ALm, mb/Ms, mb-Ms, Lc, Tme, P/Lg, relevance, wave-form complexity, spectral ratio, P wave initial motion direction, etc. The meanings of some these symbol features are listed in Table I :

TABLE I. MEANING OF CHARACTERISTIC PARAMETERS

Symbol	Meaning
Ap1	The amplitude of P wave first arrival
Asm	The maximal amplitude of the S wave group
Apm	The maxima amplitude of the P wave group
Tme	The duration of effective waveform after the time Asm occurring
ALm	The maxima amplitude of the Love wave group
mb	The body wave magnitude of the event
Ms	The Richter magnitude of the event

In order to avoid the effect of magnitude among different features, all features have been normalized to [0, 1] interval before further processing

$$x_k = (x_k - x_{\min}) / (x_{\max} - x_{\min}) \quad (6)$$

where x_k is k -th feature, $x_{\min}$, $x_{\max}$ are the minimum and maximum of the feature

B. Classification Performance Comparison

In order to derive the validity and priority of BP-Adaboost method, BP-NN and PCA-SVM method are utilized to provide comparisons. When implementing BP-NN, in order to enhance the network classification performance, double hidden layers structure is adopted, 4 nodes in each hidden layer, operating by iteration for 100 times, learning rate is 0.1, the training error is set to 10^{-5} ; And for SVM, in order to enhance the SVM classification performance, data being preprocessed by principal component analysis (PCA) (keep energy 95%); Finally for BP-Adaboost, in order to weak BP-NN performance, single hidden layer BP structure is adopted, 3 nodes in hid-den layer, operating by iterations for only 5 times, other settings are the same as the BP-NN, a total of 10 "weak classifiers" merge into a "strong classifier". Program running environment is: MATLAB R2010b, Windows 7 system.

(1) Experimental Results by Various Partitions of Dataset

Various training samples sets and testing samples sets are formed from the original dataset by partition strategy of non-overlapping, thus training sample sets and testing sample sets are completely independent. Choices are among 10% to 100%, with increment 10% (such as randomly selecting 10%(20%, 30%, ..., 90%, 100%) of original dataset as

training set, the remaining 90%(80%, 70%, ..., 10%, 100%) as testing set), each of the three classifiers are running 10 times and misclassified samples are listed in Table 2 (when training sample and testing sample partition are both 100%, mean that all sample are utilized in both training and testing). The means of misclassified samples counts in each of 10 times running, are shown in Fig. 3:

TABLE II. MISCLASSIFIED SAMPLE NUMBERS FOR VARIOUS SAMPLE PARTITIONS

Training Sample Proportion	BP-NN	Avg	PCA-SVM	Avg	BP-Adaboost	Avg
10%	13,14,9,10,28,11,18,23,38,8	17.2	4,7,5,8,6,5,20,3,5,9	7.2	0,0,0,0,0,0,0,0,0,0	0
20%	19,8,10,9,8,12,10,11,19,7	11.3	3,6,6,7,4,4,4,2,4,2	4.2	0,8,0,0,0,0,1,7,0,0	1.6
30%	6,9,8,5,4,7,8,5,6,9	6.7	2,4,7,3,7,5,5,4,5,3	4.5	0,0,0,4,0,0,5,4,0,0	1.3
40%	8,6,5,8,6,6,10,6,6,9	7.0	3,2,3,5,1,4,4,4,5,4	3.5	2,0,3,0,3,0,2,0,0,0	1.0
50%	4,6,3,13,7,4,3,2,6,6	5.4	3,1,3,3,2,3,3,2,4,3	2.7	0,3,0,0,0,5,0,1,0,0	0.9
60%	3,4,7,3,6,5,10,3,6,5	5.2	2,2,2,3,3,4,4,3,3,2	2.8	2,0,0,0,0,0,2,0,1,1	0.6
70%	4,5,4,2,3,4,2,2,6,3	3.5	2,3,1,1,3,2,1,2,2,2	1.9	0,0,2,2,0,2,1,2,0,1	1.0
80%	2,2,0,4,1,3,2,1,2,3	2.0	3,1,0,1,2,3,2,0,1,2	1.5	1,1,0,1,0,0,0,1,2,0	0.6
90%	3,0,0,3,0,0,2,1,0,2	1.1	0,0,0,2,0,0,0,0,0,1	0.3	1,0,0,0,0,0,1,0,0,0	0.2
100%	3,3,4,3,3,14,3,3,2,4	4.2	1,1,1,1,1,1,1,1,1,1	1.0	2,1,2,2,2,2,2,1,1,1	1.6

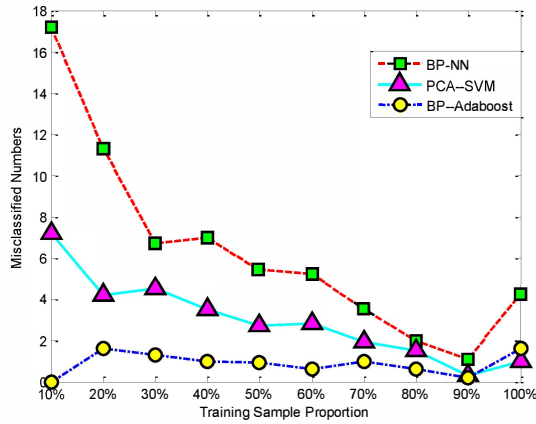


Figure 3. Average of misclassified sample numbers in 10 running for 10 partitions by 3 different classifiers

The followings can be easily to derive from figure 3:

- (i) The results of BP-Adaboost for 10 partitions of dataset exhibit rather robust; BP-NN and PCA-SVM are relative sensitive to different partitions, with the training set of sample number increase the misclassified number decreased significantly;
- (ii) If the data sample number is small, BP-Adaboost method would more reliable.

(2) Experimental Results by a Specific Partition of Dataset

TABLE III. CLASSIFICATION ACCURACY OF THREE CLASSIFIERS

Classification Algorithm	Minimum Accuracy	Maximum Accuracy	Average Accuracy
BP-NN	96.92%	98.97%	98.21%
PCA-SVM	98.46%	99.49%	99.23%
BP-Adaboost	98.97%	100%	99.49%

70% samples of dataset are randomly selected as the training set, and the remaining 30% samples as the testing set. Table 3 lists the lowest classification accuracy, the highest classification accuracy and the average classification accuracy of the 3 classifiers after running 10 times:

The three classifiers are running 10 times and each classification accuracy in 10 times running is shown in Fig. 4.

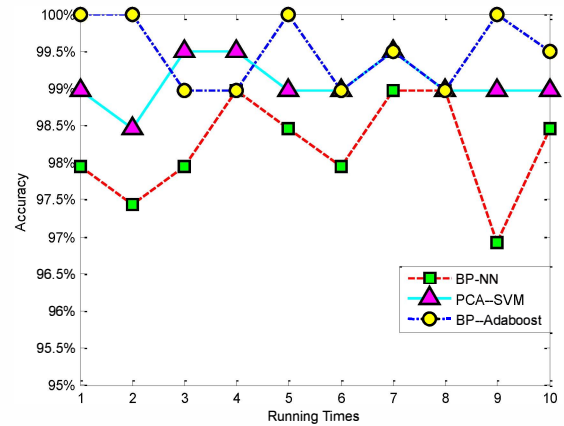


Figure 4. Classification accuracies in 10 time running for the 3 classifiers

The followings can be derived from table 3 and figure 4:

- (i) The overall classification performances of BP-Adaboost and PCA-SVM are superior to that of BP-NN. The cause may be that the BP-NN training is only based on the empirical risk minimization, and PCA-SVM needing feature selection in the first, this may eliminate for small contribution to the classification feature, and also SVM-based on structural risk minimization principle to network training, both may be the causes to result in training result is superior to the BP-NN; Although BP-Adaboost and BP-NN

(ii) BP-NN shows great instability, which may due to randomly selection of initial network weights, whilst PCA-SVM and BP-Adaboost are relatively stable.

(iii) Adaboost is considered to be by maximizing the classification of the training sample space to improve the generalization ability[10]. Above experimental results show that the Adaboost has better classification margin on the dataset if being compared with SVM.

In order to further explore the 3 classifiers' performances for earthquake and explosion recognition problem, the average classification results of 10 running results are selected as the reference., select the most close to the average classification accuracy of the results produced 3 classifier confusion matrix, the result as shown in table IV, V, VI:

TABLE IV. CONFUSION MATRIX OF BP CLASSIFICATION RESULTS

	True"1"	True"0"
Predict	TP	FP
"1"	40	3
Predict	FN	TN
"0"	1	151

TABLE V. CONFUSION MATRIX OF PCA-SVM CLASSIFICATION RESULTS

	True"1"	True"0"
Predict	TP	FP

TABLE VII. THREE KINDS OF CLASSIFIERS PERFORMANCE EVALUATION

Algorithm	P ₁	P ₂	R ₁	R ₂	F ₁	F ₂	ACC
BP-NN	93.02%	99.34%	97.56%	98.05%	95.24%	98.69%	97.95%
PCA-SVM	97.56%	99.35%	97.56%	99.35%	97.56%	95.35%	98.97%
BP-Adaboost	97.56%	100%	100%	99.35%	98.76%	99.67%	99.49%

TABLE VIII. THE AVERAGE RUNNING TIME OF THE 3 CLASSIFIERS

Algorithm	Average Running Time(s)
BP-NN	1.572477
PCA-SVM	0.353826
BP-Adaboost	6.242371

The followings can be derived from table VII and table VIII:

(i) BP-Adaboost method in Precision, Recall, F value and Accuracy is superior to the BP-NN and PCA-SVM method, and the performance has improved more than 2%. Considering BP-NN and SVM methods are strengthen form in this dissertation, the performance improvement of BP-Adaboost is more obvious.

(ii) PCA-SVM and BP-Adaboost method are relatively similar in some classification evaluation index, if we pursue real time performance, the PCA-SVM method with faster training speed could conform to our requirements, but if we demand higher prediction accuracy, the BP-Adaboost method would be more appropriate.

(iii) What we have noticed is that the dataset is unbalanced, and [9] has proved that Adaboost method is effective in unbalanced classification problem, and has a better classificationper formance generally. With regard to

	"1"	"0"
Predict	FN	TN
"1"	40	1
"0"	1	153

TABLE VI. CONFUSION MATRIX OF BP-ADABOOST CLASSIFICATION RESULTS

	True"1"	True"0"
Predict	TP	FP
"1"	41	1
Predict	FN	TN
"0"	0	153

Based on the above confusion matrix, can get the classifier performance evaluation indexes: Precision, Recall, F value and Accuracy, the corresponding computation formula is as follows (subscript 1 class for "1" (seismic), subscript 2 class for "0" (artificial explosion)) :

$$(i) \text{ Precision: } P_1 = \frac{TP}{TP + FP} \quad P_2 = \frac{TN}{TN + FN} \quad (7)$$

$$(ii) \text{ Recall: } R_1 = \frac{TP}{TP + FN} \quad R_2 = \frac{TN}{FP + TN} \quad (8)$$

$$(iii) \text{ F value: } F_1 = \frac{2 \times P_1 \times R_1}{P_1 + R_1} \quad F_2 = \frac{2 \times P_2 \times R_2}{P_2 + R_2} \quad (9)$$

$$(IV) \text{ Accuracy: } A = \frac{TP + FN}{P + N} \quad (10)$$

We caculate three kinds of classifiers performance evaluation index and the average running time. As shown in table VII and table VIII:

BP-Adaboost method, in the respect of increasing the number of "weak classifier" appropriation or enhancing the BP performance, this method can achieve 100% accuracy.

(iv) BP-Adaboost method use greedy strategy to iteratively in function space optimization specific loss function, which can effectively explain the theory of the algorithm. BP-Adaboost has good generalization capability [11] and it provides a good theoretical basis for the application of earthquakes and explosion events in the classification of seismic wave signals.

C. Other Dataset Classification Results Validation

In order to further validate the proposed methods, we have selected 35 earthquakes and 27 explosion events occurred in the Beijing-nearby-area as the dataset. 3 seismic station's data has be selected in every event, using the empirical mode decomposition(EMD) of energy ratio as the feature. Dataset size is 186 * 7 and first column works as sample label and the remaining six columns as the EMD features, selecting 70% of the data as a training set randomly, 30% of the data as a test set, and three kinds of classifiers are trained for 10 times. The result is shown in Fig. 5:

The Fig. 5 shows that the BP-Adaboost method is apparently superior to the BP-NN and PCA-SVM

performance. BP-Adaboost method has a certain robustness in earthquakes and explosion events and it can be used to obtain a better classification performance in other dataset.

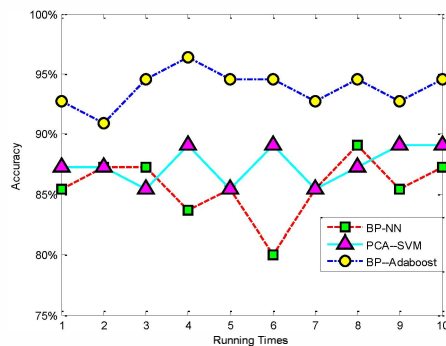


Figure 5. Classification accuracies in 10 time running for the 3 classifiers on EMD features

IV. CONCLUSION

In this paper, the BP-Adaboost method is selected for the classification of earthquake and explosion by seismic signals, and makes a performance comparison with two main classification methods: BP-NN and SVM. Experiments show that BP-Adaboost can obtain better classification results than BP-NN and SVM, in addition, BP-Adaboost method has better generalization capability, so it would be more suitable for the classification of earthquake and explosion by seismic signals.

More fresh or sophisticated methods are continually been devised, but as Manuel [12] says: do we really need so many classifiers to do research? How to find rapid and accurate classification method to identify earthquake and explosion by seismic signals is still a question worthy to study.

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